Laser Review

# DIAL Measurements of Natural and Anthropogenic Gas Emissions

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The application of a mobile DIAL (differential absorption lidar) remote sensing system in measurements of gas fluxes is discussed. UV-DIAL has been employed in the monitoring of sulphur dioxide and mercury emissions both from natural and anthropogenic sources. The concentration distributions downwind from the sources, deduced from vertical lidar scans or traverses under the plumes, were combined with wind data to determine the atmospheric fluxes of these gases. Measurements of the total flux of sulphur dioxide from the Italian volcanoes Etna, Stromboli and Vulcano were performed from an oceanographic research ship making traverses under the volcanic plumes with the lidar system sounding vertically. Some results from lidar measurements on atmospheric atomic mercury in Italian geothermal fields are also presented. Finally, some examples of remote monitoring of industrial emissions are given. The mobile DIAL system has been used for measurements on several Swedish industrial plants within the framework of a control programme commissioned by the Swedish Environmental Protection Agency.

Key Words: DIAL, Gas flux, Sulphur dioxide, Mercury

## 1. Introduction

There is an increasing need for powerful measurement techniques for natural and man-made air pollutant emissions, in order to investigate the atmospheric budget for different species, and due to more stringent regulations on industrial emissions. Optical remote sensing techniques are particularly advantageous in this context, allowing large-area monitoring and avoiding sample extraction and preparation difficulties. Long-path absorption techniques, such as differential optical absorption spectroscopy (DOAS), have been very successful during recent years and are routinely used to monitor the air quality in many polluted and clean-air areas. While DOAS can only give the mean pollutant concentration over a fixed path, three-dimensional measurements can be performed with the differential absorption lidar (DIAL) technique, which now is in operation for many pollutants. The ability to remotely measure the flux of a certain gas from different sources is particularly useful. This is achieved by combining wind data with a mapping of the concentration distribution in a vertical section downwind from the source. In this way, diffuse emissions can also be studied and the total pollutant flux can be measured. In this paper we will give some examples of the application of a mobile DIAL system in flux measurements of sulphur dioxide and mercury from different sources.

## 2. Lidar Equipment and Measurement Methodology

The mobile DIAL system used in these measurements is a flexible system designed for the detection of various species. The system has previously been described in detail[1], but several of the subsystems have since been upgraded. The laser, telescope and all electronics are contained in a Volvo F610 truck with a specially designed cargo compartment measuring $6.0 \times 2.3 \times 2.1$ m$^3$. The truck is equipped with four sturdy supporting legs which can be hydraulically lowered to ensure high stability during measurements requiring high directional accuracy in the optical system. Electric power in the field is supplied by a 20 kW diesel generator installed in a

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Table I Data for the Mobile DIAL System.

| | |
|---|---|
| | **Emitter** |
| Laser | Continuum YG682-20, Nd:YAG rep. rate 20 Hz, 6-9 ns pulse length 1200 mJ at 1064 nm |
| Dye laser | Continuum TDL60 frequency doubling/mixing 200-4500 nm, 2-150 mJ, 0.5 mrad dual-wavelength, alternate-switching |
| Wavelengths | $SO_2$: 300.02 nm (on), 299.30 nm (off) Hg: 253.652 nm, 253.665 nm |
| | **Receiver** |
| Telescope | Newtonian, 40 cm diameter, f/2.5 computer-controlled steering mirror |
| Interference filters | $SO_2$: 38% transm., 5 nm bandwidth Hg: 12%, 9 nm, or 30%, 45 nm |
| Photo detector | Thorn-EMI 9816QA, S20 cathode gain $10^7$, ramped 2-10 $\mu$s |
| Digitizer | LeCroy units 6102, TR 8818, NM8103A, 8013, 8-bit, 100-MHz |
| Computers | 2 IBM-compatible PC 386 |

small trailer towed by the truck.

The important characteristics of the updated DIAL system are summarised in Table I. The laser transmitter is a Nd:YAG-pumped dye laser with frequency doubling and/or mixing to the UV and IR regions. The dye laser is equipped with a dual-wavelength option, enabling the laser to be fired alternately at two different pre-set wavelengths. The outgoing laser beam is directed coaxially with a vertically mounted telescope and transmitted into the atmosphere via a large flat mirror in a retractable transmitting/receiving dome on the roof. A quartz window seals the dome. Stepping motors are used to turn the dome and to tilt the mirror. It is thus possible to steer the measurement direction 360 degrees horizontally and within an angle of 45 degrees vertically. An extra folding mirror can be installed on the roof to facilitate vertical measurements. An adjustable field stop is placed in the focus of the receiving Newtonian telescope, by means of which the field of view can be varied from 2 to 5 mrad. After passing through an interference filter the light is detected by a photomultiplier tube. Due to the fact that the beam is emitted coaxially with the telescope, the near-field backscattered light is very strong. To prevent detector overload and to reduce the dynamic range of the signal, the gain of the PMT is modulated. This is performed by changing the voltage difference in the dynode chain. The gain is very low in the beginning and does not reach its full value until after 2-10 $\mu$s (the rise time is variable). Due to geometrical effects of the Newtonian telescope and possible nonlinearities in the detector, the signal from the first 200 m is normally not used in the evaluation sequence.

In measurements of gas fluxes, vertical lidar scans are made downwind from all the sources. The lidar system should be placed in such a position that the complete plume can be scanned at an angle of about 90°to the wind direction. Normally, angles between 50° and 130° are adequate, whereas other angles make the flux determination very sensitive to an accurate measurement of the wind direction and to small wind shifts during the lidar measurements. Data from a vertical scan are presented as a 2D plot with the concentration value indicated by a grey-scale. The data can also be transferred to a single vertical and horizontal profile with a projection algorithm. The flux is calculated by first integrating the measured concentrations over the area of the vertical plane mapped out by the scan. This integrated value is then multiplied by the wind speed perpendicular to the plane at the height of the plume's centre of gravity. If the plume is distributed over a large height interval, it is necessary to integrate the concentration in different height intervals separately and multiply by the wind speed at the appropriate height before summing the total flux. The accuracy of the wind measurements is very important for the flux determination. The lidar system is equipped with a wind monitor mounted 5 m above the ground, but additional monitors at other heights are also used. If the position of the source of the pollutant is known, the wind direction is revealed by the position of the measured plume. Sometimes, visible plumes in or near the measurement area can be used to determine the wind field. This is done by following and timing structures in the plume moving between two lines of sight from the lidar system.

## 3. Measurements of Volcanic Gas Emissions

Measurements of the composition and flux of emitted gases and particles from active volcanoes are of great importance for the monitoring of volcanic activity, but also from an environmental point of view regarding possible effects on the earth's climate. $SO_2$ is generally the main sulphur-containing species in a high-temperature volcanic gas plume. Ash particles or condensed water in the plume can act as oxidising catalysts converting $SO_2$ into sulphates. If the sulphur in the plume remains as $SO_2$

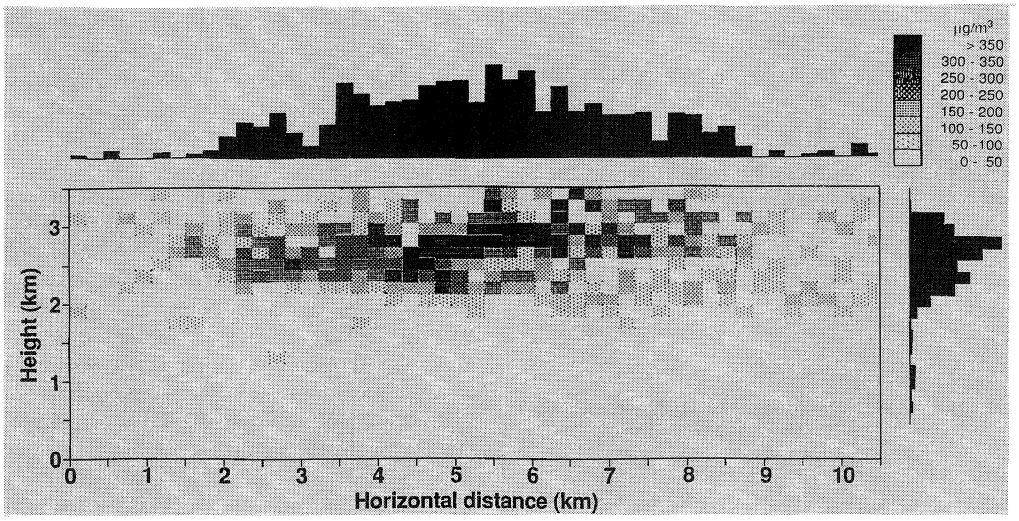



Fig. 1 $SO_2$ distribution in a cross-section of the volcanic plume from Etna, obtained from vertical DIAL measurements during a traverse under the plume.

for a long time after its dispersion into the atmosphere, and the total $SO_2$ flux is known, the flux of other chemical constituents with sufficiently long mean lifetime can also be inferred from their concentration ratio to $SO_2$.

During a field campaign in Italy the total flux of $SO_2$ from the volcanoes Etna, Stromboli and Vulcano was determined using the DIAL technique[2]. The measurements were performed with the mobile DIAL system parked on the aft deck of the Italian CNR research ship "Urania". Different measurement modes were used. At Etna and Stromboli scans through the volcanic plume were obtained in traverses under the plume, with the lidar system pointing in a fixed vertical direction. At Vulcano, with the active area at a lower altitude, vertical lidar scans through the spreading plume near the source could be made with the ship positioned in a fixed position. Fig. 1 shows an example of a measurement of the $SO_2$ plume from Etna, captured at a distance of 23 km from the source. Wind data were received from observation stations at Etna at 2500 and 3000 metres above sea level. By combining the integrated gas concentration over the plume cross-section with wind velocity data, it was possible to determine the total flux of $SO_2$. The three volcanoes were all measured within a three-day period in September 1992. We found total fluxes of about 25, 180 and 1,300 tonnes/day for Vulcano, Stromboli and Etna, respectively. A new campaign during September 1994 confirmed these results. Lidar measurements on atomic mercury were also made for the plumes from Vulcano and Stromboli, but the system sensitivity and range only allowed estimates of upper limits for the Hg fluxes.

The lidar measurements of volcanic gas fluxes are the first, to our knowledge, in which an active optical remote sensing technique has been used. During the measurements the lidar data were compared with simultaneous recording with passive DOAS and correlation spectroscopy (COSPEC) techniques using the sky radiation as the light source. A distinct difficulty with passive optical measurements, compared with active ones, is establishing the effective path through the plume. In an ideal passive measurement all detected photons are scattered above the plume and penetrate the plume in vertical beams. In a horizontally extended plume, photons can penetrate at a slanting angle before they are scattered within or below the plume. This light has a longer absorption path through the plume and will therefore indicate a larger overhead burden. If, on the other hand, the base of the plume is high enough compared with its horizontal extension, light can be scattered into the telescope without passing the plume at all. The active and passive techniques gave similar results for the low-altitude plumes from Stromboli and Vulcano, whereas the passive methods seem to overestimate the emission from Etna. The difference is dependent on the sun angle, the meteorological conditions, and the height and horizontal extension of the plume.

## 4. Measurements of Atomic Mercury of Geophysical Origin

Atomic mercury is frequently found at elevated concentrations in geothermal fields. This can constitute a pollution problem but it can also provide an interesting possibility for prospecting for geothermal energy. The contribution of the geothermal sources to the worldwide mercury flux is actually poorly investigated. Italian plans to further expand geothermal power production have initiated interests in the monitoring of these fluxes in certain regions in Italy. We have performed several field campaigns at geothermal power plants as well as in unexploited field with natural surface geothermal manifestations[3]. Highly elevated concentrations of atomic mercury were found, in contrast to earlier measurements in Iceland. Intercomparisons with the results from point monitors, based on gold amalgamation combined with flameless atomic absorption spectrometry, showed a very good agreement.

One of the advantages of using a remote sensing technique with scanning possibilities is illustrated in Fig. 2. The concentration of atomic mercury near the cooling towers of a 30-MW geothermal power plant was mapped by the lidar. The figure shows the result of several vertical and horizontal scans, as indicated by the dots. From this charting it is evident that the main part of the mercury is not emerging together with the visible water vapour plume from the top of the cooling tower, but rather it escapes from the openings in the lower part of the tower. The high levels of atmospheric mercury probably come from a process of collection and release of the mercury contained in the cooling air by the water that trickles inside the tower. Since earlier estimations of the mercury emissions were based on the mercury content in the water vapour plume, these findings necessitated a large revision of the flux values. Lidar measurements on this and similar power plants gave a mercury flux of around 25 g/h, which is similar to a typical emission from a chlor-alkali plant.

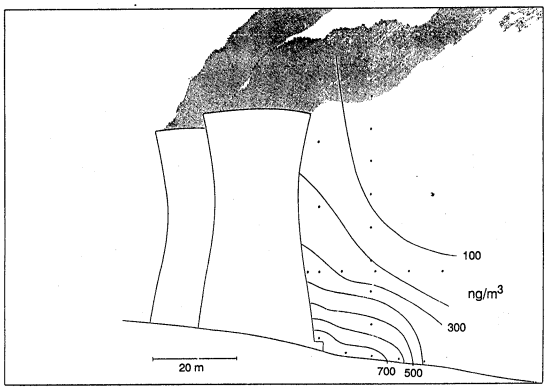



Fig. 2 Distribution of mercury in the vicinity of the cooling towers at an Italian geothermal power plant.

## 5. Measurements of Industrial Emissions

The mobile DIAL system has been employed in monitoring of industrial pollutant emissions. Measurements of total atmospheric fluxes of sulphur dioxide and mercury vapour were performed at 9 different Swedish industrial plants within the framework of a control programme commissioned by the Swedish Environmental Protection Agency[4]. $SO_2$ was monitored at six different kinds of industries. At several of these plants the flux values deduced from the lidar measurements could be compared with measurements made by the companies themselves. Especially interesting studies of the precision of the lidar flux values could be made in situations where only one or a few well-controlled sources were monitored. Direct comparisons could then be made with the flux determined from in situ $SO_2$ monitors and measurements of the gas flow inside the smokestack. These comparisons show that the values from single point sources correlate well, and indicate that lidar can determine the mean flux over some hours with an accuracy of within $\pm 10\%$ during favourable conditions. The accuracy can in some cases be worse than this, since it depends on range, wind conditions, distribution of the plume, and topography of the measurement area. Thus, it is difficult to give a general accuracy for emission values determined by a lidar system, but the sensitivity and accuracy can be estimated for each individual measurement.

A significant advantage of the lidar technique is its ability to measure the flux from several sources, extended sources and diffuse emissions. Fig. 3 shows an example of $SO_2$ lidar measurements at a metallurgical plant, which contains several different sources. The result of a vertical scan downwind from all the sources is displayed in part a) of the figure. As can be seen, at least three plumes contribute to the total flux. Part b) shows another type of representation which can be made from such a measurement. Here the concentrations are vertically integrated and the resulting vertical column content is copied onto a map of the industrial area. The direction of the scan is indicated by the horizontal axis. Together with the wind direction, this type of figure can be useful in tracking down the sources,

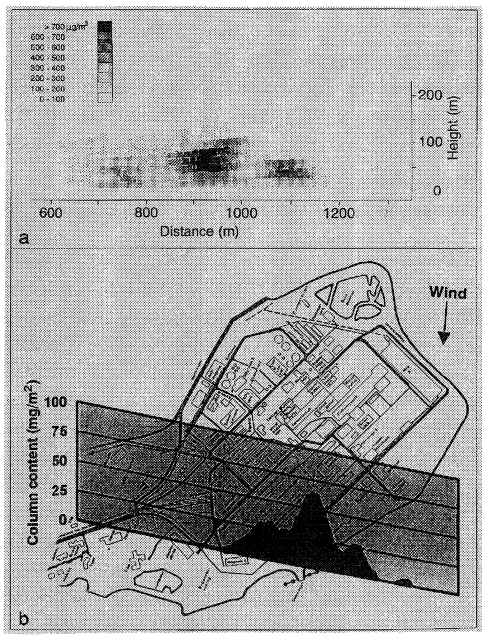



Fig. 3 $SO_2$ mapping downwind from a Swedish metallurgical plant. a) $SO_2$ distribution in a vertical plane. b) $SO_2$ vertical column content, copied onto a map of the industrial area.

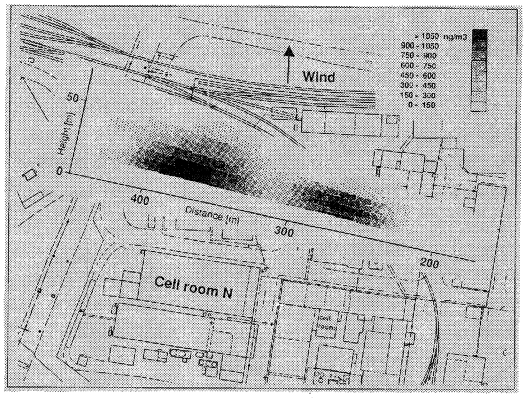



Fig. 4 Distribution of mercury in a vertical plane scanned by the DIAL system at a Swedish chlor-alkali plant. The horizontal direction for the vertical scan is indicated by the Distance axis.

especially if measurements are made in various directions during different wind conditions. A comparison was made of the total $SO_2$ flux deduced from the lidar measurements and from in situ measurements made by the company. The latter value is calculated by summing up emission values from six different locations, which are continuously monitored, and values from manual measurements made irregularly at some additional points. The lidar gave on average 30-50% higher values for the total $SO_2$ emission. This difference is probably due to diffuse emissions, which are not registered in the measurements made by the company.

Mercury emissions were also monitored at six different plants. One of the major Hg emission sources is the chlor-alkali industry, where a flowing cathode of metallic mercury is often used in an electrolytic cell. Fig. 4 shows the results of a lidar mercury measurement at such a plant. The largest emission source here is the main electrolytic cell building (cell room N), from which mercury is ventilated through openings along the roof. The spreading Hg plume from this building can be seen in the figure, although part of the plume may be undetected since the lower angles in the vertical scan were limited by a nearby building. Closer to the lidar system, a second smaller plume can also be seen, which was not expected. The source of this was found to be a smaller cell house used intermittently, where some construction work was carried out during the measurements. This activity obviously released a large amount of mercury and contributed substantially to the total Hg flux. Estimates of the Hg flux to the atmosphere were made by the company only at the main cell house. This estimate involved measuring the concentration inside the building and determining the air flow from different subsections. A comparison with these data shows that the lidar measurements gave a total flux twice as high as the company's own estimate. A similar study at another chlor-alkali plant gave a lidar value of four times the earlier estimation, although no 'extra' plume was detected here. This substantial difference is surprising, since it is not likely that the lidar overestimates the emission.

## 6. Conclusions

The data presented here show that the lidar technique can be a very useful method of monitoring both natural and anthropogenic gas emissions. Our measurements on volcanic emissions indicate that DIAL can measure the overhead burden more correctly than passive techniques using scattered sun light, which often require correction for scattering within or below the plume. The DIAL technique also gives the height and vertical extension of the plume which are important parameters in the determination of the appropriate wind velocity. One general advantage of the lidar technique is the possibility of rapidly measuring the total emission from

several sources, including diffuse emission, in an area without any preknowledge of the exact position of the different sources. This ability is quite unique and often makes lidar measurements more cost-effective than the numerous measurements needed with point monitors or long-path absorption techniques to achieve the same result. The measurements of industrial emissions presented here were performed within the framework of a control program during announced visits to different plants. However, since lidar is a remote sensing technique capable of detecting plumes at distances of up to a few km, it can also be employed successfully in unannounced inspections.

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